Oscar Hernandez

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Education

Bard College at Simon's Rock — **Great Barrington, MA**
B.A. Mathematics and Computer Science — *– Spring 2017*
A.A. Liberal Arts and Sciences — *Fall 2013 – Spring 2015*

Experience

ARKS — **Spark Tank, Riverside, CA**
Co-founder & Computer Engineer — *January 2019 –*

- Engage in the online YC Startup School.
- Design intelligent autonomous robotic system to automate internal warehouse logistics.

Zap Crunch — **ExCITE, Riverside, CA**
Freelance Full-Stack Web Developer — *February 2018 –*

- Build and maintain custom web applications using the Django.py, Flask.py, and Express.js frameworks and APIs from Shopify, Brandboom, WordPress, Twilio, and Google Maps.
- Build and maintain websites with Hugo, Jekyll, or Drupal, implementing the Netlify Content Management System when necessary.

Independent Research & Consulting — **Spark Tank, Riverside, CA**
Cross-disciplinary scientist and engineer — *January 2018 –*

- Researching machine learning, specifically transfer learning and the capacity of convolutional neural networks.
- Develop web apps/sites and manage information technology for artists and academics, including Columbia University's American Assembly.
- Former machine learning scientist at VNenergy, a startup in the Los Angeles Cleantech Energy Innovators Program, from August 2018 until its closing in December 2018.

Success Academy Charter Schools — **Harlem, NY**
5th Grade Associate STEM Teacher — *July 2017 – December 2017*

- Taught 5th Grade History, ELA, Math, Earth Science courses, and independently designed, implemented, and taught the school's entire 5th-8th Grade Computer Science elective curricula from scratch.

Bard College at Simon's Rock
2x Summer Research Intern, Research Assistant, Thesis-ing Senior — *May 2015 – June 2017*

- Conducted research on economic game dynamics, and wrote manuscript on computational game complexity [Her+18].
- Published research [DHW16; Dra+18; Has+17] on theoretical computer science and biomimetic electronics, and awarded Honors for senior thesis on discrete mathematics [HW17].

Computer Skills

General: Python, JavaScript, Shell, SQL, Haskell, Prolog, Ruby, Java, C, C++;

Design: HTML, CSS, PHP, Markdown, LaTeX, Blender, Maya, Unity3D;

Web: Django, Rails, Flask, Express, Koa, Meteor, React, Angular, Bootstrap, Jekyll, Hugo;

Applications: Git+GitHub, Nginx, Apache, DigitalOcean, Heroku, AWS, GCP, Netlify, Wordpress.

Publications

[Dra+18] Patrick Dragon et al. "Constructing de Bruijn sequences with Co-Lexicographic Order: The k-ary Grandmama Sequence". In: *European Journal of Combinatorics* 72 (Aug. 2018), pp. 1–11. URL: https://www.sciencedirect.com/science/article/pii/S0195669818300696.

[HW17] Oscar Hernandez and Aaron Williams. "Constructing Maximum-Rotating de Bruijn Sequences Using Necklace Trees". thesis. Great Barrington, MA: Bard College at Simon's Rock, May 2017. URL: http://digitalcommons.bard.edu/sr-theses/1120/.

[Has+17] Harold Hastings et al. "Dynamics of Biomimetic Electronic Artificial Neural Networks". In: *Proceedings of the 4th International Conference on Applications in Nonlinear Dynamics*. Ed. by Visarath In, Patrick Longhini, and Antonio Palacios. Vol. 6. Lecture Notes in Networks and Systems. Denver, CO: Springer International Publishing AG, Mar. 2017, pp. 195–207. URL: https://link.springer.com/chapter/10.1007/978-3-319-52621-8_18.

[DHW16] Patrick Dragon, Oscar Hernandez, and Aaron Williams. "The Grandmama de Bruijn Sequence for Binary Strings". In: *LATIN 2016: Theoretical Informatics*. Ed. by Evangelos Kranakis, Gonzalo Navarro, and Edgar Chávez. Vol. 9644. Lecture Notes in Computer Science. Ensenada, Mexico: Springer-Verlag Berlin Heidelberg, Apr. 2016, pp. 347–361. URL: https://link.springer.com/chapter/10.1007/978-3-662-49529-2_26.

Manuscripts

[HI19] O. Hernandez and M. Izbicki. "Zero Shot Sentiment Analysis on Tweets in Any Language". Presented at the SoCal NLP Symposium 2019 at USC. Sept. 2019.

[Her+18] O. Hernandez et al. "Spinning Around and Around: Turnstile Puzzles are PSPACE-Complete". Apr. 2018.

Interests

Current: Humanities, STEM, Category Theory, Artificial Intelligence, Free Software/Culture

Prospective: Perl, Google Apps Script, Native Android Apps, Categorical Physics, Systems Theory